

Clustering using K-Medoids with Jaccard Distance

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1. Overview

In this assignment, we implemented K-Medoids clustering on three text datasets (KOS, NIPS, and Enron) using Jaccard distance as the similarity measure. Since the data is in Bag-of-Words format, we converted it into a Set-of-Words representation to make the Jaccard distance applicable.

2. Parameters Used

Distance Measure: Jaccard Distance

Clustering Algorithm: K-Medoids

Number of iterations (max_iter): 10

Number of runs per K (for stability): 3

Range of K values explored:

- Small range: $K = 2$ to 12
- Extended range: $K = 2$ to 100

Sampling used:

- KOS: 2000 documents
- NIPS: approximately 1200 documents
- Enron: approximately 4000 documents

Sampling was necessary to reduce computational cost and make clustering feasible.

3. Implementation Details

Each document was represented as a set of unique word IDs, effectively converting the Bag-of-Words representation into a Set-of-Words model.

Jaccard distance was used to measure dissimilarity between documents, defined as:

$$d(A, B) = 1 - (|A \cap B| / |A \cup B|)$$

K-Medoids clustering was implemented from scratch with the following steps:

- Random initialization of medoids
- Assignment of each document to the closest medoid
- Updating medoids by minimizing intra-cluster distance
- Sampling within clusters (up to 50 documents) to speed up medoid computation

To improve robustness, clustering was run multiple times for each value of K, and the result with the lowest cost was selected.

4. Time Taken

The runtime varied depending on the dataset size and value of K.

Approximate time per run (per value of K):

- KOS dataset: 18–22 seconds
- NIPS dataset: 10–15 seconds
- Enron dataset: 25–40 seconds

Running clustering for multiple values of K (from 2 to 12) took a few minutes per dataset.

5. Space Requirements

Memory usage was primarily determined by:

- Storage of document sets (dictionary of sets)
- Cluster assignments during execution

The implementation is memory-efficient as it avoids storing dense matrices and relies on sparse set representations.

6. Observations

The clustering cost decreases as the number of clusters (K) increases. However, the rate of decrease slows down beyond a certain point, indicating diminishing returns.

From the elbow analysis:

- KOS dataset: optimal $K \approx 12$
- NIPS dataset: optimal $K \approx 10$
- Enron dataset: optimal $K \approx 12$

In all three datasets, no sharp elbow was observed. Instead, a gradual flattening region was identified, indicating that moderate values of K provide a good balance.

7. Challenges Faced

- High computational cost of Jaccard distance
 - Large size of datasets, especially Enron
 - Sensitivity of K-Medoids to initial medoid selection
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8. Optimizations Used

To address performance challenges, the following optimizations were applied:

- Random sampling of documents
- Sampling within clusters during medoid update
- Running the algorithm multiple times and selecting the best result

These optimizations significantly improved runtime while maintaining clustering quality.

9. Conclusion

K-Medoids clustering with Jaccard distance is well-suited for text data represented as sets. While increasing the number of clusters reduces clustering cost, the improvement becomes

marginal beyond a certain point. Therefore, moderate values of K provide an effective trade-off between clustering quality and model complexity.